

GENERALIZED MOMENT RETRIEVAL · 实验全景与真实性追溯大表

数据快照：2026-07-23 23:46 CST

评估数据集：Standard validation (465 queries: 255 pos / 210 null)

追溯标准：100% JSON 文件物理追溯； 代表日志双重审计通过

骨干架构	实验方案 / 变体	AUROC ↑	REJ- F1@0.4 ↑	MAP ↑	MR@5 (OR @3) ↑	MR+@5 (OR @3) ↑	G-MIOU@3 ↑	追溯 JSON 文件	日志
Moment-DETR 骨干模型族									
Moment-DETR	Moment-DETR Strict GMR (Base)	70.95%	61.22%	8.93%	13.85%	3.17%	30.78%	md_gmr_b128/best_joint...metrics.json	✓
	Moment HieA2M Parent	73.38%	40.40%	9.16%	15.45%	1.33%	15.54%	md_hiea2m_b128_rerun...metrics.json	✓
	HieA2M-DGQC + Independent Zero	72.62%	69.28%	9.16%	15.45%	1.33%	39.77%	stage2_zero/best_joint...metrics.json	✓
	+ Zero (seed2023)	72.86%	1.86%	9.16%	3.08%	seed2023/stage2_zero...metrics.json	✗		
	+ Zero (seed2024)	59.53%	0.00%	9.16%	2.65%	seed2024/stage2_zero...metrics.json	✗		
	+ Zero (seed2025)	73.61%	0.94%	9.16%	2.87%	seed2025/stage2_zero...metrics.json	✗		
	+ Zero (seed2023_posw4)	63.46%	0.00%	9.16%	2.65%	seed2023_posw4/stage2_zero...metrics.json	✗		
	Moment + Quality + Dual (Stage B)	71.80%	53.40%	7.48%	12.34%	1.67%	23.53%	moment/md_quality_dual/best_joint...metrics.j...	✓
	Base GMR + Direct Top-3 (Dedup K=3)	6.48%	9.25%	seed2023/stage4_5_selection/learned_selector...	✗				
	Base GMR + Hard-NMS (Dedup K=3)	6.48%	9.25%	seed2023/stage4_5_selection/learned_selector...	✗				
	Base GMR + Learned Dedup (K=3)	6.96%	9.25%	seed2023/stage4_5_selection/learned_selector...	✗				
	HieA2M Rerun v2 + Direct Top-3 (Dedup K=3)	9.16%	13.85%	md_hiea2m_b128_rerun.../dedup_ablation_summar...	✗				
	HieA2M Rerun v2 + Hard-NMS (Dedup K=3)	9.16%	13.85%	md_hiea2m_b128_rerun.../dedup_ablation_summar...	✗				
	HieA2M Rerun v2 + Learned Dedup (K=3)	9.16%	13.85%	md_hiea2m_b128_rerun.../dedup_ablation_summar...	✗				
	MD HieA2M Release seed2018	72.94%	67.54%	9.70%	15.07%	0.83%	35.31%	runs/md_hiea2m_release_seed2018...metrics.json	✗
	MD HieA2M Release seed2023	72.89%	68.63%	10.34%	15.53%	1.17%	36.90%	runs/md_hiea2m_release_seed2023...metrics.json	✗
	MD HieA2M Release seed2024	72.75%	67.54%	9.55%	14.10%	1.06%	35.01%	runs/md_hiea2m_release_seed2024...metrics.json	✗
	MD GMR Continue lr5e-6 seed2023	72.55%	66.52%	9.18%	14.08%	0.83%	34.43%	runs/md_gmr_continue_lr5e6_seed2023...metrics...	✗
EaTR 骨干模型族									
EaTR	EaTR Strict GMR (Base)	71.67%	39.35%	8.02%	12.30%	2.93%	16.82%	eatr_gmr_strict/best_joint...metrics.json	✓
	EaTR Strict GMR (Best mAP)	71.67%	39.35%	8.02%	16.82%	eatr_gmr_strict/best_map...metrics.json	✗		
	EaTR Plain (Frozen Parent)	69.59%	0.00%	8.73%	12.85%	1.37%	3.51%	frozen_parent/eatr_plain_b128...metrics.json	✗
	+ Quality (EaTR Quality)	71.98%	44.31%	8.24%	12.69%	4.17%	19.13%	eatr_quality/best_joint...metrics.json	✓
	+ Quality (Best mAP)	71.79%	30.88%	8.37%	13.14%	eatr_quality/best_map...metrics.json	✗		

	+ Dual Grounding (EaTR Dual)	72.05%	48.12%	8.06%	11.91%	1.31%	21.10%	eatr_dual/best_joint...metrics.json		✔	
	+ Dual Grounding (Best mAP)	72.05%	48.12%	8.06%	21.10%	eatr_dual/best_map...metrics.json		✘			
	+ Counter (EaTR Counter)	73.31%	29.79%	6.16%	9.46%	0.93%		12.15%	eatr_counter/best_joint...metrics.json		✔
	+ Counter (Best mAP)	73.16%	16.60%	8.66%	8.04%	eatr_counter/best_map...metrics.json		✘			
	完整 HieA2M-DGQC Full	72.65%	22.66%	7.08%	9.89%	1.61%		9.56%	eatr_hiea2m/best_joint...metrics.json		✔
	完整 HieA2M-DGQC Full (Best mAP)	72.71%	18.62%	8.72%	8.42%	eatr_hiea2m/best_map...metrics.json		✘			
	EaTR + Quality + Dual (Stage B)	71.90%	41.64%	7.85%	12.57%	3.87%		17.57%	eatr/eatr_quality_dual/best_joint...metrics.j...		✔
	Base GMR + Direct Top-3 (Dedup K=3)	6.96%	1.81%	eatr_gmr_strict/dedup_ablation...json		✘					
	Base GMR + Hard-NMS (Dedup K=3)	6.96%	1.81%	eatr_gmr_strict/dedup_ablation...json		✘					
	Base GMR + Learned Dedup (K=3)	6.95%	1.81%	eatr_gmr_strict/dedup_ablation...json		✘					
	Canonical EaTR seed2023	69.10%	69.95%	4.00%	45.50%	canonical/eatr...json		✘			
	Canonical Restart EaTR seed2023	69.91%	0.00%	8.38%	3.69%	canonical_b128_restart/eatr...json		✘			
	formal EaTR seed2023 (无 GMR)	69.24%	0.00%	7.79%	11.52%	0.70%		3.39%	formal/eatr...json	✘	
	formal EaTR GMR seed2023	70.12%	68.66%	7.37%	11.40%	0.76%		42.98%	formal/eatr_gmr...json	✘	
Flash-VTG 骨干模型族											
Flash-VTG	Flash-VTG GMR (Base, seed2023)	72.07%	62.59%	26.53%	35.72%	12.96%		35.39%	flash_vtg_gmr/best_joint...metrics.json		✔
	Flash-VTG Plain (Base)	59.93%	41.06%	22.00%	32.41%	8.37%		20.57%	flash_vtg_plain/best_joint...metrics.json		✔
	Flash-VTG GMR Release Anchor	73.95%	62.53%	26.01%	34.88%	12.93%		33.93%	release_gmr_val/strict_gmr...json		✘
	+ Quality (Flash Quality)	73.95%	62.53%	26.67%	37.61%	14.80%		34.03%	flash_vtg_gmr_quality...metrics.json		✔
	+ Zero (Flash Zero)	74.29%	60.98%	26.01%	34.88%	12.93%		31.66%	flash_vtg_gmr_zero...metrics.json		✔
	+ Quality + Zero	74.13%	61.12%	26.67%	37.61%	14.80%		32.86%	flash_vtg_gmr_quality_zero...metrics.json		✔
	Flash-VTG GMR seed2024	74.15%	64.15%	26.57%	34.47%	12.69%		34.90%	seed2024/flash_vtg_gmr...metrics.json		✘
	Base GMR + Direct Top-3 (Dedup K=3)	23.49%	8.56%	flash_vtg_gmr/dedup_ablation...json		✘					
	Base GMR + Hard-NMS (Dedup K=3)	23.14%	8.56%	flash_vtg_gmr/dedup_ablation...json		✘					
	Base GMR + Cluster Fusion (K=3)	24.28%	8.56%	flash_vtg_gmr/dedup_ablation...json		✘					
QD-DETR 骨干模型族											
QD-DETR	QD Strict GMR (Base)	72.40%	3.74%	7.03%	9.10%	0.00%		3.14%	qd_detr_gmr/best_joint...metrics.json		✔
	QD + Quality (bsz32)	72.50%	70.24%	7.56%	10.80%	0.67%		42.36%	qd_quality/best_joint...metrics.json		✔
	QD + Dual (bsz32)	72.38%	67.98%	6.61%	10.36%	0.17%		38.54%	qd_dual/best_joint...metrics.json		✔
	QD + Counter (bsz32)	72.33%	26.72%	6.76%	9.76%	0.00%		9.75%	qd_counter/best_joint...metrics.json		✔
	QD + HieA2M (bsz32)	72.55%	0.00%	5.83%	8.83%	0.00%		2.24%	qd_hiea2m/best_joint...metrics.json		✔

	QD Strict GMR (Best mAP)	72.40%	3.74%	7.03%	3.14%	qd_detr_gmr/best_map...metrics.json	✖		
	QD + Quality (Best mAP)	72.50%	70.24%	7.56%	42.36%	qd_quality/best_map...metrics.json	✖		
	QD + Counter (Best mAP)	72.72%	0.00%	7.03%	2.34%	qd_counter/best_map...metrics.json	✖		
	QD + HieA2M (Best mAP)	72.40%	0.00%	6.80%	2.19%	qd_hiea2m/best_map...metrics.json	✖		
	Continued Control	72.02%	65.96%	6.91%	9.95%	0.67%	35.23%	qd_fair_ablation/.../continued_control...json	✔
	+ Quality	72.10%	63.35%	6.54%	9.68%	0.67%	31.75%	qd_fair_ablation/.../quality...json	✔
	+ Dual	72.40%	66.38%	7.15%	10.33%	0.00%	35.03%	qd_fair_ablation/.../dual...json	✔
	+ Quality + Dual	72.74%	70.26%	6.27%	9.32%	0.00%	42.04%	qd_fair_ablation/.../quality_dual...json	✔
	QD Dual Best (e48)	72.38%	67.98%	6.61%	10.36%	0.17%	38.54%	strict_bsz32/qd_detr/.../qd_dual/best_map...j...	✔
	QD + Quality + Dual (Stage B)	72.74%	70.26%	6.27%	9.32%	0.00%	42.04%	qd/qd_quality_dual/best_joint...json	✔
	Base GMR + Direct Top-3 (Dedup K=3)	5.84%	0.00%	qd_detr_gmr/dedup_ablation...json	✖				
	Base GMR + Hard-NMS (Dedup K=3)	5.82%	0.00%	qd_detr_gmr/dedup_ablation...json	✖				
	Base GMR + Cluster Fusion (K=3)	5.84%	0.00%	qd_detr_gmr/dedup_ablation...json	✖				
	QD Quality + Direct Top-3 (Dedup K=3)	6.36%	0.67%	qd_quality/dedup_ablation...json	✖				
	QD Quality + Hard-NMS (Dedup K=3)	6.32%	0.67%	qd_quality/dedup_ablation...json	✖				
	QD Quality + Cluster Fusion (K=3)	6.36%	0.67%	qd_quality/dedup_ablation...json	✖				
	QD Dual + Direct Top-3 (Dedup K=3)	5.47%	0.00%	qd_dual/dedup_ablation...json	✖				
	QD Dual + Hard-NMS (Dedup K=3)	5.47%	0.00%	qd_dual/dedup_ablation...json	✖				
	QD Dual + Cluster Fusion (K=3)	5.44%	0.00%	qd_dual/dedup_ablation...json	✖				
	Canonical QD-DETR seed2023	53.32%	42.02%	0.07%	17.22%	canonical/qd_detr...metrics.json	✖		
	Canonical Restart QD-DETR seed2023	53.32%	42.02%	0.07%	17.22%	canonical_b128_restart/qd_detr...metrics.json	✖		
	formal QD-DETR seed2023 (无 GMR)	71.99%	0.00%	7.29%	10.28%	0.50%	2.83%	formal/qd_detr...metrics.json	✖
	formal QD-DETR GMR seed2023	61.03%	70.83%	7.73%	13.24%	0.72%	46.25%	formal/qd_detr_gmr...metrics.json	✖
CG-DETR 骨干模型族									
CG-DETR	CG Strict GMR (Base)	59.69%	0.00%	4.84%	7.64%	0.33%	1.87%	cg_detr_gmr/best_joint...metrics.json	✔
	CG + Quality (bsz32)	59.75%	0.00%	4.95%	7.08%	0.72%	1.79%	cg_quality/best_joint...metrics.json	✔
	CG + Counter (bsz32)	59.71%	0.00%	4.58%	6.67%	0.39%	1.77%	cg_counter/best_joint...metrics.json	✔
	CG + HieA2M (bsz32)	59.68%	0.00%	4.65%	6.81%	0.39%	1.86%	cg_hiea2m/best_joint...metrics.json	✔
	CG + Phrase (bsz32)	59.72%	0.00%	4.76%	7.19%	0.22%	1.86%	cg_phrase/best_joint...metrics.json	✔
	CG Strict GMR (Best mAP)	59.69%	0.00%	4.84%	1.87%	cg_detr_gmr/best_map...metrics.json	✖		

	CG + Quality (Best mAP)	59.75%	0.00%	4.95%	1.79%	cg_quality/best_map...metrics.json	✖		
	CG + Counter (Best mAP)	59.62%	0.00%	4.59%	1.66%	cg_counter/best_map...metrics.json	✖		
	CG + HieA2M (Best mAP)	59.68%	0.00%	4.65%	1.86%	cg_hiea2m/best_map...metrics.json	✖		
	CG + Phrase (Best mAP)	59.72%	0.00%	4.76%	1.86%	cg_phrase/best_map...metrics.json	✖		
	Base GMR + Direct Top-3 (Dedup K=3)	3.50%	0.00%	cg_detr_gmr/dedup_ablation...json	✖				
	Base GMR + Learned Dedup (K=3)	3.50%	0.00%	cg_detr_gmr/dedup_ablation...json	✖				
	CG Quality + Direct Top-3 (Dedup K=3)	3.57%	0.17%	cg_quality/dedup_ablation...json	✖				
	CG Quality + Learned Dedup (K=3)	3.57%	0.17%	cg_quality/dedup_ablation...json	✖				
	Canonical CG-DETR seed2023	43.34%	0.00%	0.01%	1.55%	canonical/cg_detr...metrics.json	✖		
	Canonical Restart CG-DETR seed2023	43.34%	0.00%	0.01%	1.55%	canonical_b128_restart/cg_detr...metrics.json	✖		
	formal CG-DETR seed2023	40.37%	0.00%	4.61%	7.46%	0.43%	2.12%	formal/cg_detr...metrics.json	✖
	formal Restart CG-DETR seed2023	40.31%	0.00%	4.94%	7.45%	0.61%	1.91%	formal_b128_restart/cg_detr...metrics.json	✖

XX-GMR (Base) 基准参考行

高亮绿色

代表指标优于对应的 **XX-GMR Base** 基线

✓

代表物理日志双重追溯通过

✖

代表暂无物理日志，仅有 JSON 物理追溯

代表实验仍处于微调训练进行中